Swiggy SQL Project

SELECT * FROM swiggy_data;

Data Cleaning

Check for Null Values in Each Column

```
SELECT
        SUM(CASE WHEN State IS NULL THEN 1 ELSE 0 END) AS null_state,
        SUM(CASE WHEN City IS NULL THEN 1 ELSE 0 END) AS null_city,
        SUM(CASE WHEN Order_Date IS NULL THEN 1 ELSE 0 END) AS null_order_date,
        SUM(CASE WHEN Restaurant_Name IS NULL THEN 1 ELSE 0 END) AS null_restaurant,
        SUM(CASE WHEN Location IS NULL THEN 1 ELSE 0 END) AS null_location,
        SUM(CASE WHEN Category IS NULL THEN 1 ELSE 0 END) AS null_category,
        SUM(CASE WHEN Dish_Name IS NULL THEN 1 ELSE 0 END) AS null_dish,
        SUM(CASE WHEN Price_INR IS NULL THEN 1 ELSE 0 END) AS null_price,
        SUM(CASE WHEN Rating IS NULL THEN 1 ELSE 0 END) AS null_rating,
        SUM(CASE WHEN Rating_Count IS NULL THEN 1 ELSE 0 END) AS null_rating_count
FROM swiggy_data;
```

Blank/Empty String Check

```
SELECT *
FROM swiggy_data
WHERE
State = ' ' OR City = ' ' OR Restaurant_Name = ' ' OR Location = ' ' OR Category = ' ' OR
Dish_Name= ‘ ';
```

Duplicate Detection

```
SELECT
State, City, order_date, restaurant_name, location, category,
dish_name, price_INR, rating, rating_count, count (*) AS CNT
FROM swiggy_data
GROUP BY
State, City, order_date, restaurant_name, location, category,
```

```
dish_name, price_INR, rating, rating_count
HAVING count (*)>1;
```

Delete Duplication

```
WITH CTE AS (
	SELECT *, ROW_NUMBER () OVER (
	PARTITION BY State, City, order_date, restaurant_name, location, category,
dish_name, price_INR, rating, rating_count
ORDER BY (SELECT NULL)
) AS rn
FROM swiggy_data
)
DELETE FROM CTE WHERE rn>1;
```

Data Preparation

CREATING SCHEMA

```
CREATE TABLE dim_date (
	date_id INT IDENTITY (1,1) PRIMARY KEY,
	Full_Date DATE,
	Year INT,
	Month INT,
	Month_Name VARCHAR (20),
	Quarter INT,
	Day INT,
	Week INT
);
```

--dim_location

```
CREATE TABLE dim_location (
	location_id INT IDENTITY (1,1) PRIMARY KEY,
	State VARCHAR (100),
	City VARCHAR (100),
	Location VARCHAR (200)
);
```

--dim_restaurant

```
CREATE TABLE dim_restaurant (
	restaurant_id INT IDENTITY (1,1) PRIMARY KEY,
```

```
	Restaurant_Name VARCHAR (200)
);

--dim_category
CREATE TABLE dim_category (
	category_id INT IDENTITY (1,1) PRIMARY KEY,
	Category VARCHAR (200)
);

--dim_dish
CREATE TABLE dim_dish (
	dish_id INT IDENTITY (1,1) PRIMARY KEY,
	Dish_Name VARCHAR (200)
);



--FACT TABLE
CREATE TABLE fact_swiggy_orders (
	order_id INT IDENTITY(1,1) PRIMARY KEY,

	date_id INT,
	Price_INR DECIMAL(10,2),
	Rating DECIMAL(4,2),
	Rating_Count INT,

	location_id INT,
	restaurant_id INT,
	category_id INT,
	dish_id INT,

	FOREIGN KEY (date_id) REFERENCES dim_date(date_id),
	FOREIGN KEY (location_id) REFERENCES dim_location(location_id),
	FOREIGN KEY (restaurant_id) REFERENCES dim_restaurant(restaurant_id),
	FOREIGN KEY (category_id) REFERENCES dim_category(category_id),
	FOREIGN KEY (dish_id) REFERENCES dim_dish(dish_id)
);

SELECT * FROM fact_swiggy_orders;
```

```
--INSERT DATA IN TABLES
--dim_date
INSERT INTO dim_date(Full_Date, Year, Month, Month_Name, Quarter, Day, Week)
SELECT DISTINCT
        Order_Date,
        YEAR(Order_Date),
        MONTH(Order_Date),
        DATENAME(MONTH, Order_Date),
        DATEPART(QUARTER, Order_Date),
        DAY(Order_Date),
        DATEPART(WEEK, Order_Date)
FROM swiggy_data
WHERE Order_Date IS NOT NULL;

SELECT * FROM dim_date;

--dim_location
INSERT INTO dim_location (State, City, Location)
SELECT DISTINCT
        State,
        City,
        Location
FROM swiggy_data;

SELECT * FROM dim_location;

--dim_restaurant
INSERT INTO dim_restaurant (Restaurant_Name)
SELECT DISTINCT
        Restaurant_Name
FROM swiggy_data;

SELECT * FROM dim_restaurant;

--dim_category
INSERT INTO dim_category (Category)
SELECT DISTINCT
        Category
FROM swiggy_data;

SELECT * FROM dim_category;
```

```
--dim_dish
INSERT INTO dim_dish (Dish_Name)
SELECT DISTINCT
        Dish_Name
FROM swiggy_data;

SELECT * FROM dim_dish;



--fact_table
INSERT INTO fact_swiggy_orders
(
        date_id,
        Price_INR,
        Rating,
        Rating_Count,
        location_id,
        restaurant_id,
        category_id,
        dish_id
)
SELECT

        dd.date_id,
        s.Price_INR,
        s.Rating,
        s.Rating_Count,

        dl.location_id,
        dr.restaurant_id,
        dc.category_id,
        dsh.dish_id

FROM swiggy_data s

JOIN dim_date dd ON
        dd.Full_Date = s.Order_Date

JOIN dim_location dl ON
        dl.State = s.State
AND dl.City = s.City
```

```
AND dl.Location = s.Location

JOIN dim_restaurant dr ON
        dr.Restaurant_Name = s.Restaurant_Name

JOIN dim_category dc ON
        dc.Category = s.Category

JOIN dim_dish dsh ON
        dsh.Dish_Name = s.Dish_Name;
```

SELECT * FROM fact_swiggy_orders;

```
SELECT * FROM fact_swiggy_orders f
JOIN dim_date d ON f. date_id = d. date_id
JOIN dim_location l ON f. location_id = l. location_id
JOIN dim_restaurant r ON f. restaurant_id = r. restaurant_id
JOIN dim_category c ON f. category_id = c. category_id
JOIN dim_dish di ON f. dish_id = di. dish_id;
```

KPI's

Total Orders

```
SELECT COUNT (order_id) AS Total_Orders
FROM fact_swiggy_orders;
```

Solution / Output:

	Total_Orders
1	197401

Total Revenue (INR Million)

```
SELECT
FORMAT (SUM (CONVERT (FLOAT, price_INR))/1000000, 'N2') + 'INR Million'
```

```
AS Total_Revenue
FROM fact_swiggy_orders;
```

Solution / Output:

	Total_Revenue
1	53.00INR Million

Average Dish Price

```
SELECT
FORMAT (AVG (CONVERT (FLOAT, price_INR)), 'N2') + 'INR'
AS Avg_Dish_Price
FROM fact_swiggy_orders;
```

Solution / Output:

	Avg_Dish_Price
1	268.50INR

Average Rating

```
SELECT
AVG(Rating) AS Avg_Rating
FROM fact_swiggy_orders;
```

Solution / Output:

	Avg_Rating
1	4.341577

Deep-Dive Business Analysis

Monthly order trends

```
SELECT
d.year,
d. month,
d. month_name,
count (*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f. date_id = d. date_id
GROUP BY
d.year,
d. month,
d. month_name;
```

Solution / Output:

	year	month	month_name	Total_Orders
1	2025	1	January	25393
2	2025	2	February	23291
3	2025	3	March	24400
4	2025	4	April	24584
5	2025	5	May	25188
6	2025	6	June	24382
7	2025	7	July	24936
8	2025	8	August	25227

Quarterly order trends

```
SELECT
d.year,
d. quarter,
COUNT (*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f. date_id = d. date_id
GROUP BY
d.year,
d. quarter
ORDER BY COUNT (*) DESC;
```

Solution / Output:

	year	quarter	Total_Orders
1	2025	2	74154
2	2025	1	73084
3	2025	3	50163

Year-wise growth

```
SELECT
d.year,
COUNT (*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f. date_id = d. date_id
GROUP BY
d.year
ORDER BY COUNT (*) DESC;
```

Solution / Output:

	year	Total_Orders
1	2025	197401

Day-of-week patterns

```
SELECT
        DATENAME (WEEKDAY, d. full_date) AS day_name,
        COUNT (*) AS Total_Orders
        FROM fact_swiggy_orders f
        JOIN dim_date d ON f. date_id = d. date_id
        GROUP BY DATENAME (WEEKDAY, d. full_date), DATEPART (WEEKDAY, d. full_date)
        ORDER BY DATEPART (WEEKDAY, d. full_date);
```

	city	Total_Orders
1	Bengaluru	20072
2	Mumbai	10507
3	Hyderabad	10308
4	Jaipur	10285
5	Lucknow	10192
6	New Delhi	10191
7	Ahmedabad	10175
8	Chandigarh	10060
9	Kolkata	10044
10	Chennai	10042

Solution / Output:

	day_name	Total_Orders
1	Sunday	28469
2	Monday	27568
3	Tuesday	27413
4	Wednesday	28284
5	Thursday	28450
6	Friday	28284
7	Saturday	28933

Top 10 cities by order volume

```
SELECT TOP 10
l.city,
COUNT (*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_location l ON f. location_id = l. location_id
GROUP BY
l.city
ORDER BY COUNT (*) DESC;
```

Solution / Output:

	city	Total_Orders
1	Bengaluru	20072
2	Mumbai	10507
3	Hyderabad	10308
4	Jaipur	10285
5	Lucknow	10192
6	New Delhi	10191
7	Ahmedabad	10175
8	Chandigarh	10060
9	Kolkata	10044
10	Chennai	10042

Revenue contribution by states

```
SELECT
l. state,
SUM (f. price_INR) AS Total_Revenue
FROM fact_swiggy_orders f
JOIN dim_location l ON f. location_id = l. location_id
GROUP BY l. state
ORDER BY Total_Revenue DESC;
```

Solution / Output:

	state	Total_Revenue
1	Karnataka	5455887.73
2	Uttar Pradesh	3117359.65
3	Telangana	3021656.62
4	Maharashtra	3015573.35
5	Delhi	2829180.60
6	Gujarat	2815536.27
7	Punjab	2804991.82
8	West Bengal	2662213.76
9	Tamil Nadu	2642594.63
10	Rajasthan	2502833.61
11	Madhya Pradesh	1969962.77

Top 10 restaurants by orders

```
SELECT TOP 10
r.restaurant_name,
COUNT (*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_restaurant r ON f. restaurant_id = r. restaurant_id
GROUP BY
r.restaurant_name
ORDER BY COUNT (*) DESC;
```

Solution / Output:

	restaurant_name	Total_Orders
1	McDonald's	13528
2	KFC	12957
3	Burger King	7115
4	Pizza Hut	6529
5	Domino's Pizza	5489
6	LunchBox - Meals and Thalis	4700
7	Baskin Robbins - Ice Cream Desserts	4197
8	Faasos - Wraps, Rolls & Shawarma	3256
9	Olio - The Wood Fired Pizzeria	3239
10	The Good Bowl	2665

Top categories (Indian, Chinese, etc.)

```
SELECT TOP 10
c.category,
COUNT (*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_category c ON f. category_id = c. category_id
GROUP BY
c.category
ORDER BY COUNT (*) DESC;
```

Solution / Output:

	category	Total_Orders
1	Recommended	24097
2	Desserts	3582
3	Main Course	2983
4	Beverages	2682
5	BURGERS	2538
6	Sweets	1954
7	McSaver Combos (2 Pc Meals)	1884
8	Exclusive Deals (Save upto 40%)	1717
9	Starters	1692
10	ROLLS	1652

Most ordered dishes

```
SELECT
d. dish_name,
COUNT (*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_dish d ON f. dish_id = d. dish_id
GROUP BY
d.dish_name
ORDER BY COUNT (*) DESC;
```

Solution / Output:

	dish_name	Total_Orders
1	Veg Fried Rice	321
2	Choco Lava Cake	303
3	Jeera Rice	265
4	Paneer Butter Masala	262
5	French Fries	248
6	Chicken Sausage	230
7	Chicken Fried Rice	228
8	Butter Naan	218
9	Margherita Pizza	203
10	Green Salad	197

Cuisine performance -> Orders + Avg Rating

```
SELECT
        c.category,
        COUNT (*) AS Total_Orders,
        AVG (f. rating) AS Avg_Rating
FROM fact_swiggy_orders f
JOIN dim_category c ON f. category_id = c. category_id
GROUP BY c. category
ORDER BY Total_Orders DESC;
```

Solution / Output:

	category	Total_Orders	Avg_Rating
1	Recommended	24097	4.321782
2	Desserts	3582	4.371635
3	Main Course	2983	4.310191
4	Beverages	2682	4.368717
5	BURGERS	2538	4.324940
6	Sweets	1954	4.455527
7	McSaver Combos (2 Pc Meals)	1884	4.413800
8	Exclusive Deals (Save upto 40%)	1717	4.348806
9	Starters	1692	4.304846
10	ROLLS	1652	4.246368

Total Orders By Price Range

```
SELECT
        CASE
                WHEN CONVERT(FLOAT, price_INR) < 100 THEN 'Under 100'
                WHEN CONVERT(FLOAT, price_INR) BETWEEN 100 AND 199 THEN '100-199'
                WHEN CONVERT(FLOAT, price_INR) BETWEEN 200 AND 299 THEN '200-299'
                WHEN CONVERT(FLOAT, price_INR) BETWEEN 300 AND 499 THEN '300-499'
                ELSE '500+'
        END AS price_range,
        COUNT(*) AS Total_Orders
FROM fact_swiggy_orders
GROUP BY
        CASE
                WHEN CONVERT(FLOAT, price_INR) < 100 THEN 'Under 100'
                WHEN CONVERT(FLOAT, price_INR) BETWEEN 100 AND 199 THEN '100-199'
                WHEN CONVERT(FLOAT, price_INR) BETWEEN 200 AND 299 THEN '200-299'
                WHEN CONVERT(FLOAT, price_INR) BETWEEN 300 AND 499 THEN '300-499'
                ELSE '500+'
        END
ORDER BY Total_Orders DESC;
```

Solution / Output:

	price_range	Total_Orders
1	100-199	56189
2	200-299	54567
3	300-499	43758
4	Under 100	26795
5	500+	16092

Rating Count Distribution (1-5)

```
SELECT
        rating,
        COUNT (*) AS Rating_Count
FROM fact_swiggy_orders
GROUP BY rating
ORDER BY Rating_Count DESC;
```

Solution / Output:

	rating	Rating_Count
1	4.40	85642
2	4.30	13698
3	4.60	10840
4	4.50	9946
5	5.00	9401
6	4.70	9089
7	4.80	8809
8	4.20	8214
9	4.10	7619
10	4.90	5713